

Digital Forensic Investigation Lifecycle using Python

Python Program:

```
import datetime

def investigation_report(case_id, evidence_list):
    stages = {
        "Identification": f"Incident reported for case {case_id}.  
Devices/logs identified for review.",
        "Collection": f"{len(evidence_list)} item(s) collected: {'  
' + '.join(evidence_list)}",
        "Preservation": "All items hashed (SHA-256) and stored in a  
write-protected evidence folder.",
        "Analysis": "Log files and file metadata examined for indicators of  
compromise.",
        "Reporting": "Findings compiled into a structured forensic report for  
legal review."
    }

    print(f"=== Investigation Lifecycle: Case {case_id} ===")
    print(f"Generated: {datetime.datetime.now()}\n")

    for stage, detail in stages.items():
        print(f"[{stage}]")
        print(f"{detail}\n")

investigation_report("CASE-2026-014", ["laptop_disk_image.dd", "router_traf  
fic.pcap", "email_headers.txt"])
```

Sample Output:

```
=== Investigation Lifecycle: Case CASE-2026-014 ===
Generated: 2026-07-16 05:43:16.508220
```

```
[Identification]
Incident reported for case CASE-2026-014. Devices/logs identified for
review.
```

```
[Collection]
3 item(s) collected: laptop_disk_image.dd, router_traffic.pcap,
email_headers.txt
```

```
[Preservation]
All items hashed (SHA-256) and stored in a write-protected evidence
folder.
```

```
[Analysis]
Log files and file metadata examined for indicators of compromise.
```

```
[Reporting]
Findings compiled into a structured forensic report for legal review.
```

Explanation:

This Python program demonstrates the digital forensic investigation lifecycle. It generates a structured report covering the five key stages of a forensic investigation: Identification, Collection, Preservation, Analysis, and Reporting. The program records the case ID, lists the collected

evidence, emphasizes evidence preservation using SHA-256 hashing and write protection, summarizes the analysis process, and concludes with report preparation. This illustrates the standard workflow followed during digital forensic investigations.